

Do Not Miss Architectural Distortion

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See the slide presentation [here](#).

The mammographic imaging features of architectural distortion (AD) are linear spiculations that radiate from a central point within the tissue, appearing as delicate linear strands extending outward (Fig 1). At US, AD may manifest as an irregular mass with associated distortion of the surrounding parenchymal tissue, such as straightening or bending of the Cooper ligaments (Fig 2). Focal retraction or distortion of the breast tissue that creates an irregular contour represents another distinct feature seen at MRI as an enhancing or nonenhancing finding.

AD is seen in 4% of breast cancers and found on 6% of abnormal screening mammograms. Postsurgical scars are by far the most common cause of AD. A study by Sickles, performed before the advent of digital breast tomosynthesis (DBT), highlighted the prevalence of AD on over 80% of mammograms obtained up to 6 months after surgical biopsy, with its appearance decreasing over time. Benign causes of AD can include radial scars and complex sclerosing lesions, sclerosing adenosis, and fat necrosis. The most common malignant causes of AD are invasive ductal carcinoma and invasive lobular carcinoma, although AD is not the most common manifestation of these entities. Ductal carcinoma in situ can also manifest with mammographic

AD in 2%–10% of patients. In cases where a biopsy of architectural distortion reveals a nonmalignant result, the most common finding is a radial scar. One study showed an upgrade rate to malignancy for AD with atypia of 28% and for AD without atypia of 2%, emphasizing the importance of assessing concordance after a biopsy.

DBT aids in identifying AD at mammography and increases AD detection by nearly twofold compared with two-dimensional mammography. The positive predictive value for suspicious AD detected at DBT is 43%–50.7%. DBT reduces the superimposition of overlapping structures, making AD more apparent and easier to detect, and its multiplanar images enable precise localization.

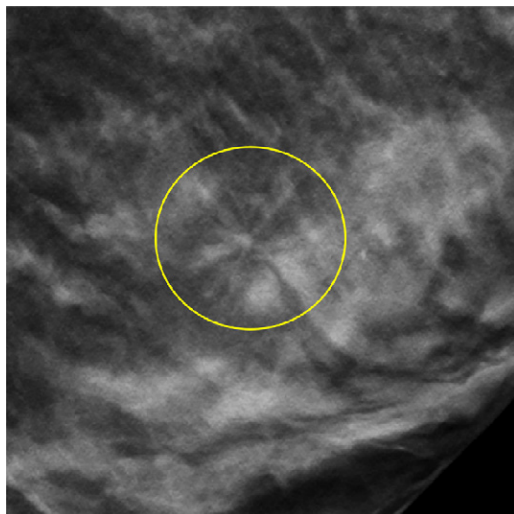


Figure 1. AD in a 68-year-old patient. Screening mammographic image on a DBT plane depicts AD (circle). Biopsy results indicated a radial scar with associated atypia.

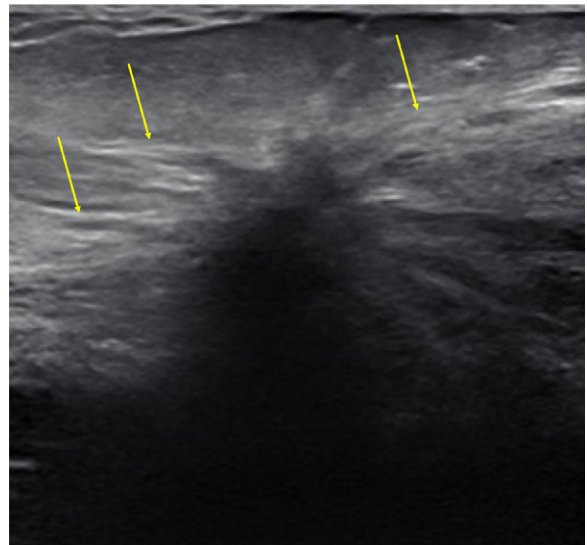


Figure 2. Architectural distortion in a 35-year-old patient who presented with a palpable left breast mass. Diagnostic US image demonstrates an irregular, not parallel, hypoechoic mass with spiculated margins and straightening of the Cooper ligaments (arrows). The mass was biopsied twice, with pathologic analysis showing a benign fibroepithelial lesion and pseudoductal stromal hyperplasia (PASH). The benign pathologic results were considered discordant to the suspicious US findings, and the patient was referred for surgical consultation. Surgical excision was recommended, and a benign radial scar without associated atypia or malignancy was diagnosed after excision. Radial scar is considered a concordant result.



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Abbreviations: AD = architectural distortion, DBT = digital breast tomosynthesis

TEACHING POINTS

- Recognizing architectural distortion, which is present on 6% of screening mammograms and characterized by linear spiculations without a visible mass, is crucial for diagnosing breast cancer.
- Architectural distortion, which can be most commonly caused by postsurgical scars, encompasses a spectrum of causes, from benign lesions such as radial scars to malignant tumors such as invasive ductal carcinoma.
- Subtle distortion, which is pronounced at DBT, is notorious for being a missed sign of breast cancer and necessitates a systematic correlation of multimodality imaging findings with clinical and pathologic findings.
- DBT has been shown to have better sensitivity for subtle architectural distortion when compared with that of two-dimensional mammography.

There are several pitfalls associated with identification of AD. Heterogeneous or extremely dense breast tissue may limit AD visibility at mammography, as the overlapping tissue may obscure AD. Further, AD that appears on only one imaging view may occur due to the orientation of the AD relative to the imaging plane and may lead to underestimation of its significance.

The accompanying slide presentation provides multimodal examples of AD with pathologic correlation and recommended management. The aim is to help the radiologist understand the differential diagnosis of AD, suggest suitable management, and correlate radiologic findings with the results of pathologic analysis. AD requires a thorough clinical and imaging evaluation.

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Suggested Readings

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